AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

1.Create an EC2 Windows Instance in AWS

SHABHARIVAASUN K R

CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,

Coimbatore.

Objective

EC2 (Elastic Compute Cloud) is a core serviceprovided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

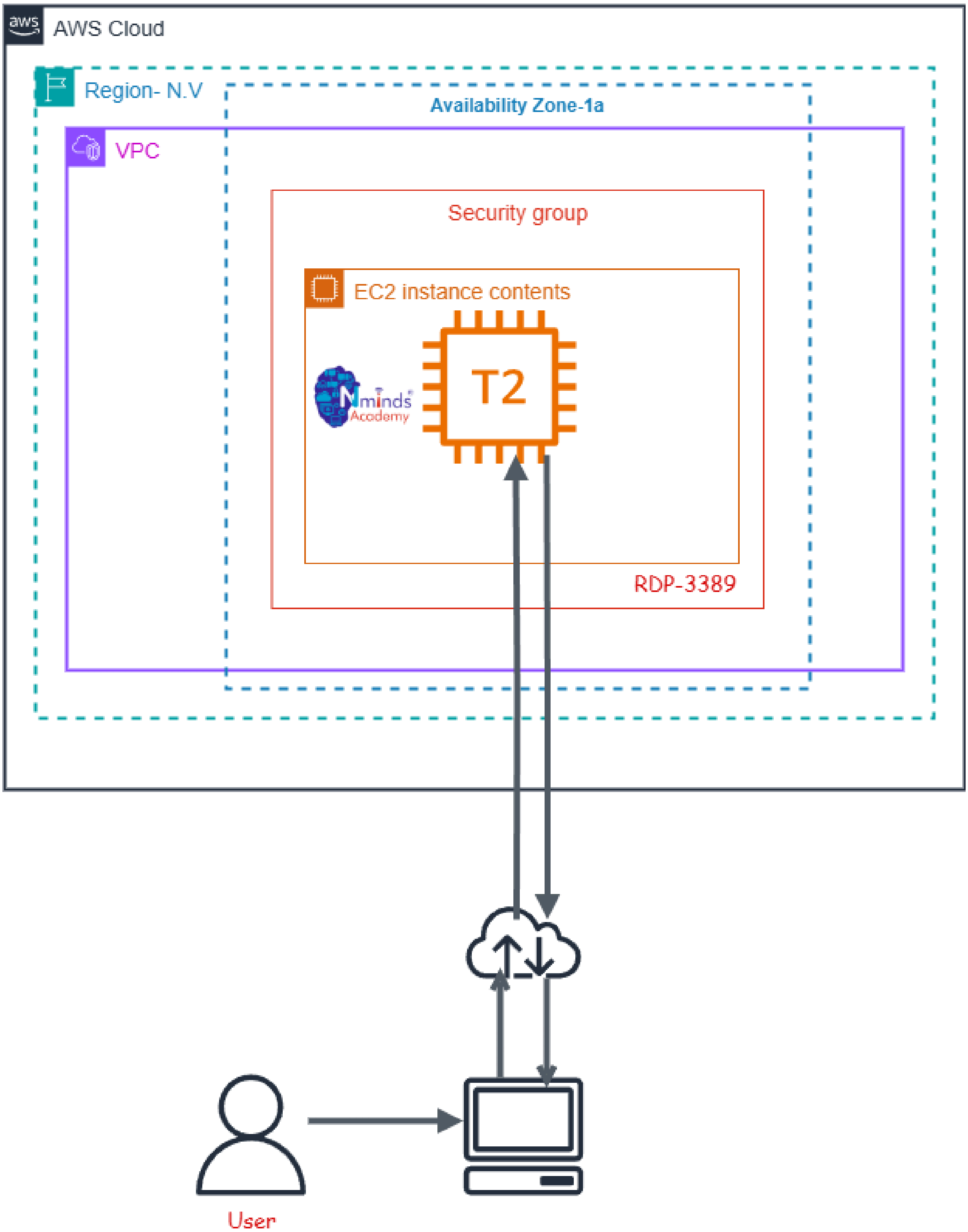
Use Cases for EC2 Windows Instances:

- Web and Application Hosting: Run IIS web servers or other Windows-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Windows instances, using RDS or EC2 for more control.
- Remote Desktop Applications: Set up RDS (Remote Desktop Services) for remote access to Windows desktops in the cloud.
- Development and Testing: Create development environments with Windows OS to test .NET or other Windows-based applications.

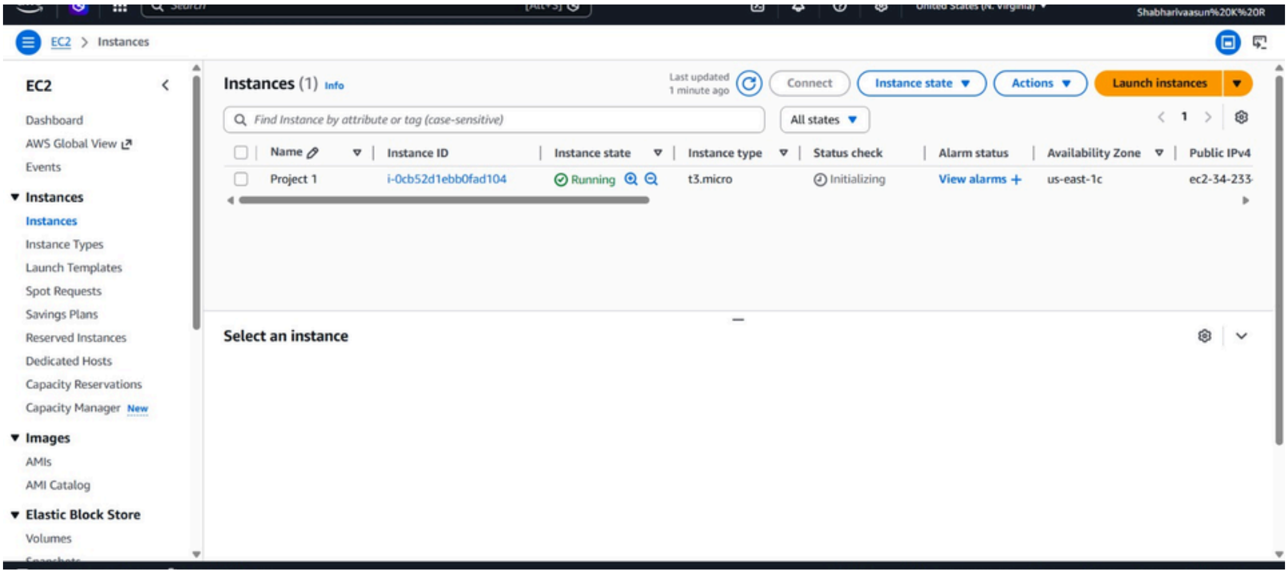
Used Services in AWS:

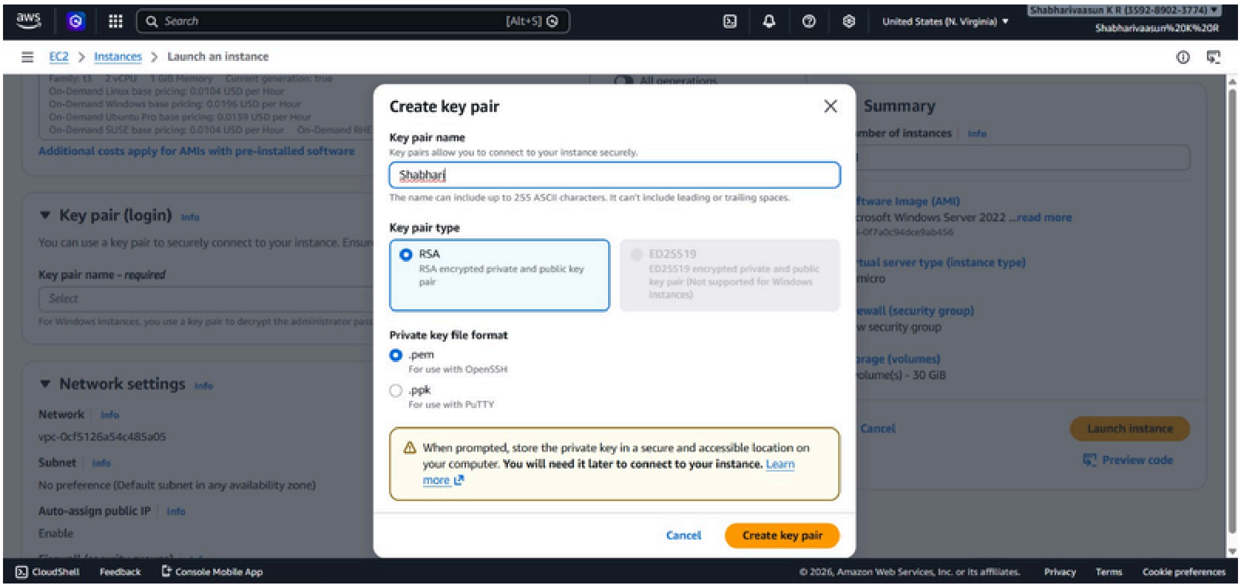
Amazon EC2

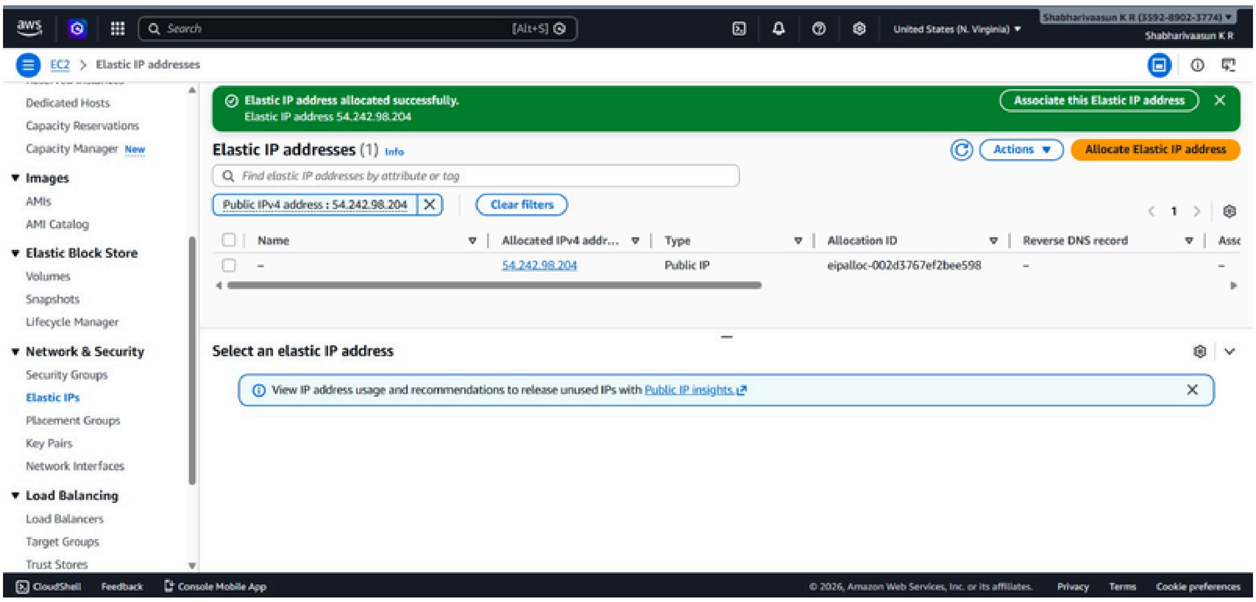
Topology


AWS Cloud
Region- N.V
Availability Zone-1a
VPC
Security group
EC2 instance contents
T2
Nminds Academy
RDP-3389
User

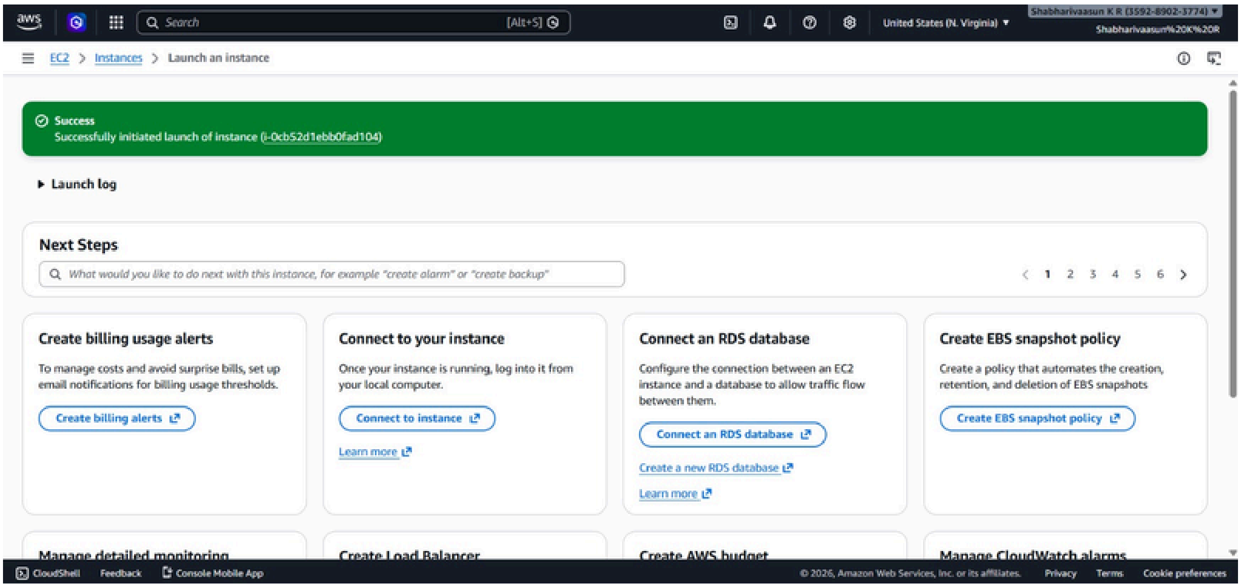
Execution Tasks:







Execution Tasks:



AWS Solution Architect Training with
AWS Cloud Practitioner Global CertificationTraining

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

2.Configure IIS Webserver on Windows Server in AWS

Windows

SHABHARIVAASUN K R

CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,
Coimbatore.

Objective

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

IIS (Internet Information Services) is a web server software created by Microsoft for hosting websites and web applications on Windows Server operating systems. It provides a platform for building, deploying, and managing web-based applications. IIS supports multiple protocols, including HTTP, HTTPS, FTP, and more, and is tightly integrated with the Windows OS.

Common Use Cases of IIS:

1. Hosting Websites: IIS is often used by businesses and developers to host websites and web applications, particularly those built with ASP.NET.
2. Web Applications: For applications that require robust back-end support, IIS provides features like application pooling and security for web services (e.g., SOAP or RESTful APIs).
3. Secure File Sharing: IIS can be configured as an FTP server to securely share files between different locations.
4. Application Hosting in Enterprises: Many enterprise applications built on Microsoft technologies rely on IIS for deployment, such as SharePoint, Exchange Server, or Dynamics CRM.
5. API Hosting: IIS is commonly used to host APIs (e.g., RESTful services) built with .NET or other technologies.

Used Services in AWS

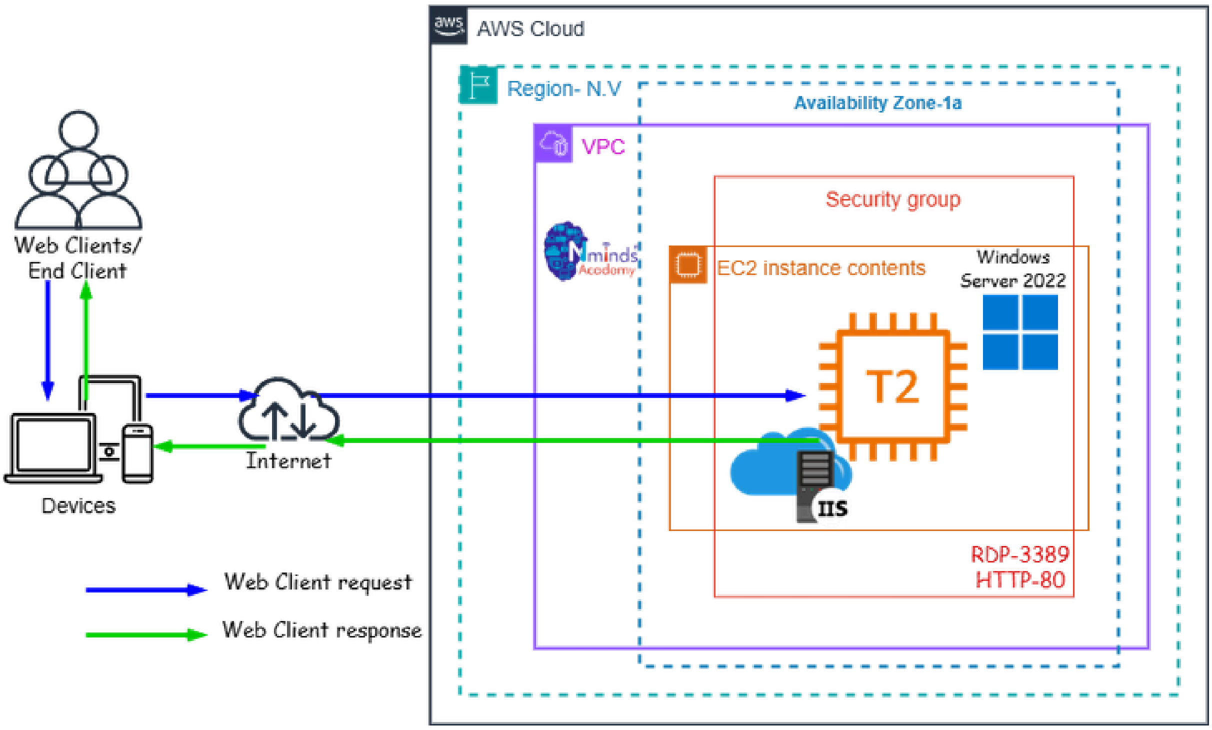
Used Services in Windows Server

IIS

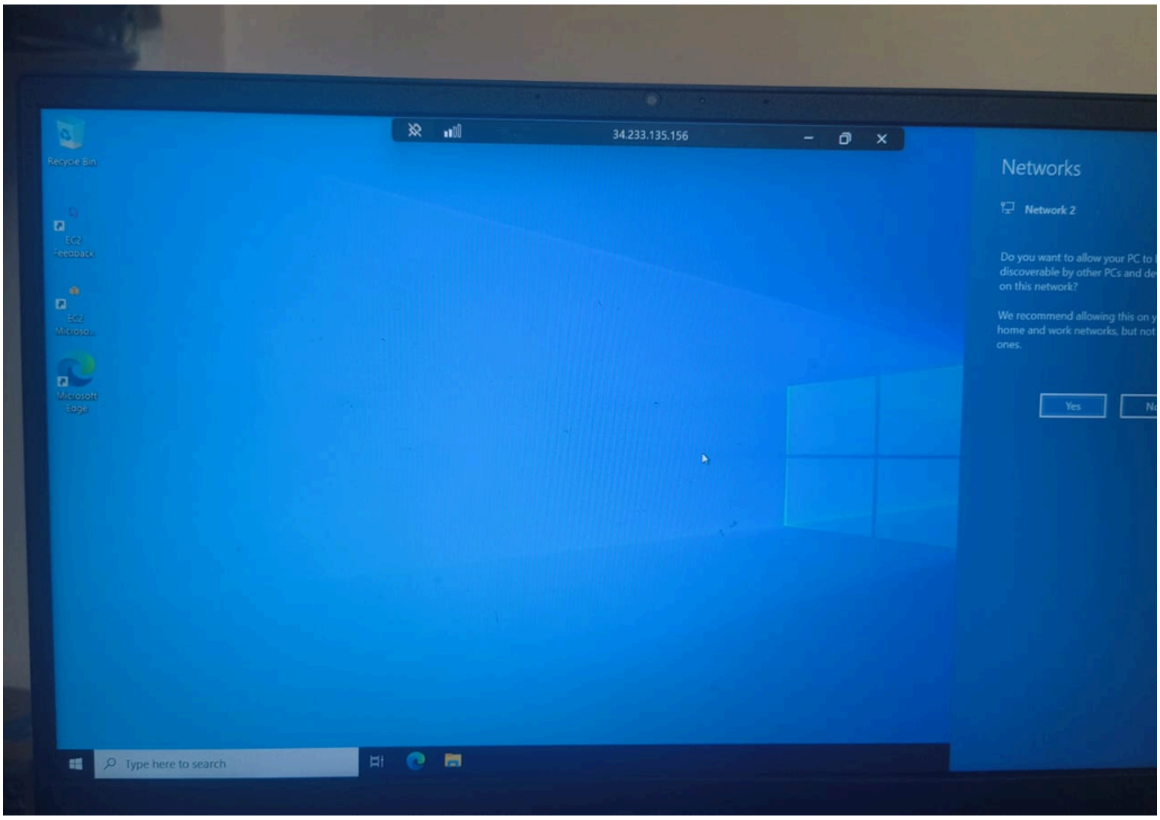
IIS- Internet Information Services

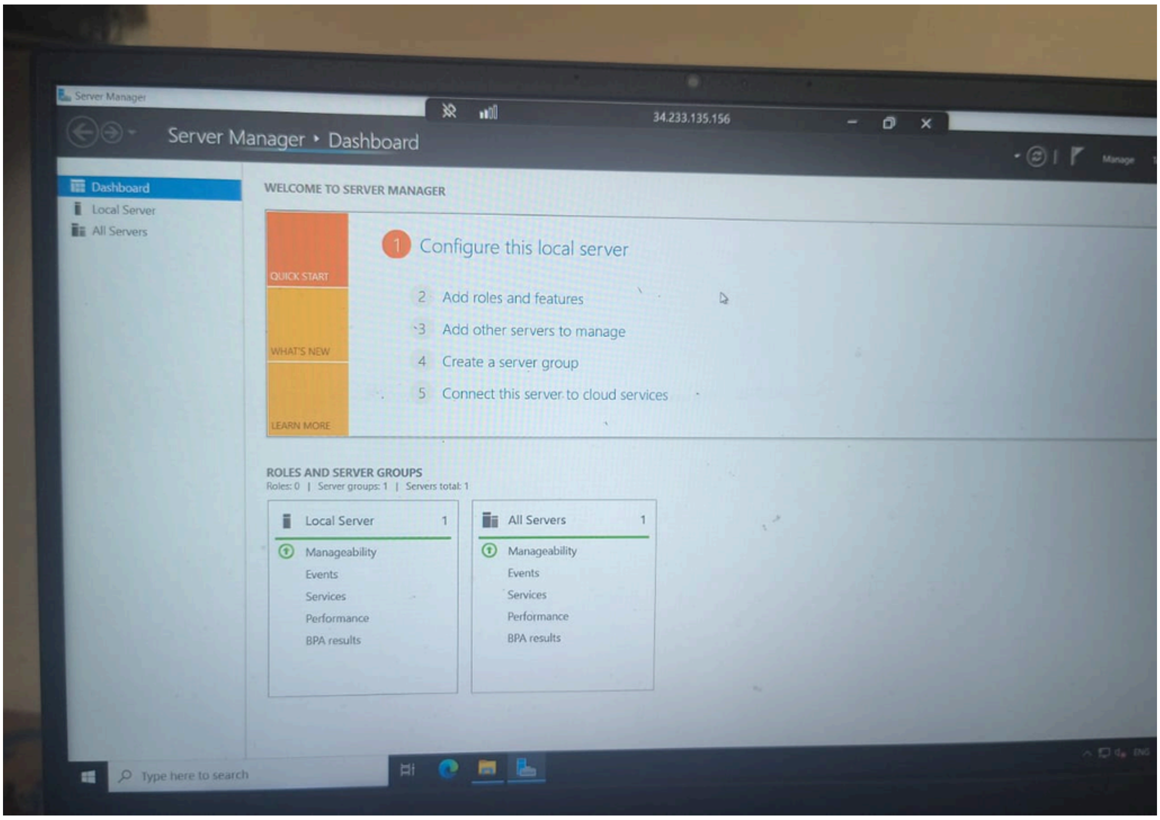
Topology

How to Configure the IIS web server in Windows Server with AWS EC2

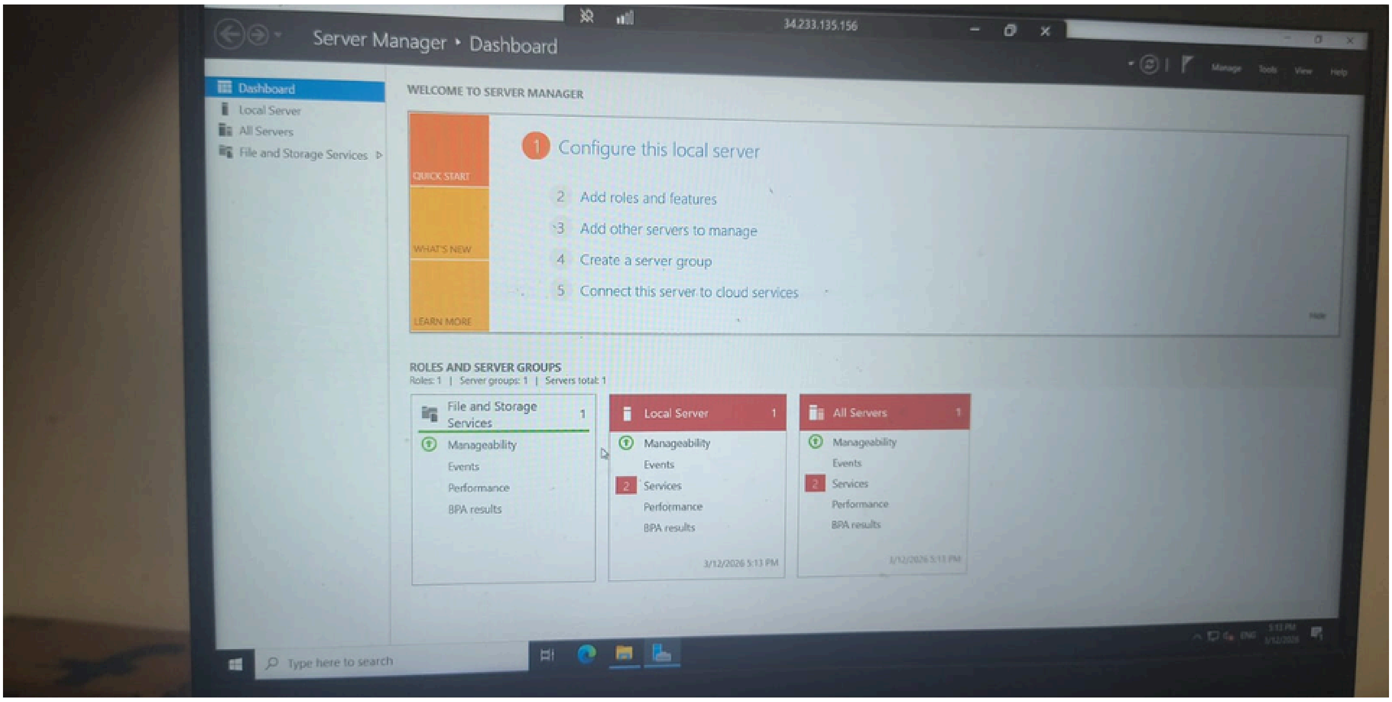


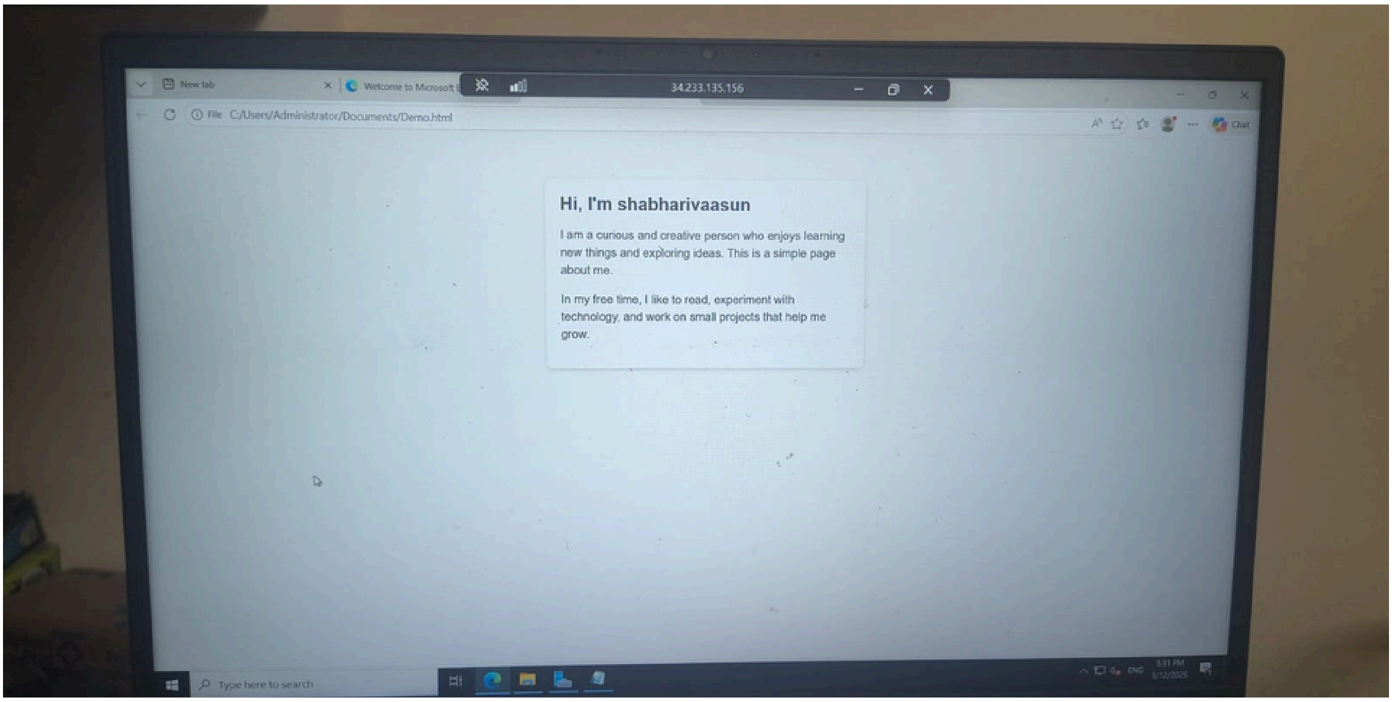
Execution Tasks:





Execution Tasks:





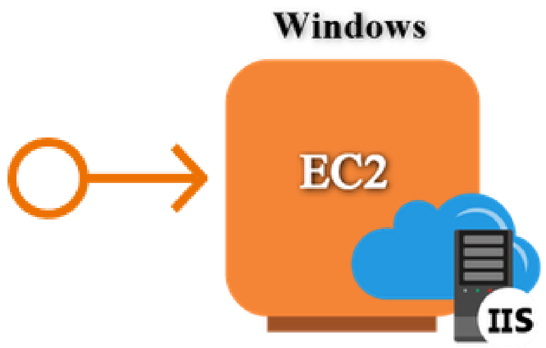
AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

3.Configure Elastic IP Address to Windows WebServer in AWS



SHABHARIVAASUN K R

CSE-Artificial Intelligence & Machine Learning, Sri Eshwar College of Engineering, Coimbatore.

Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

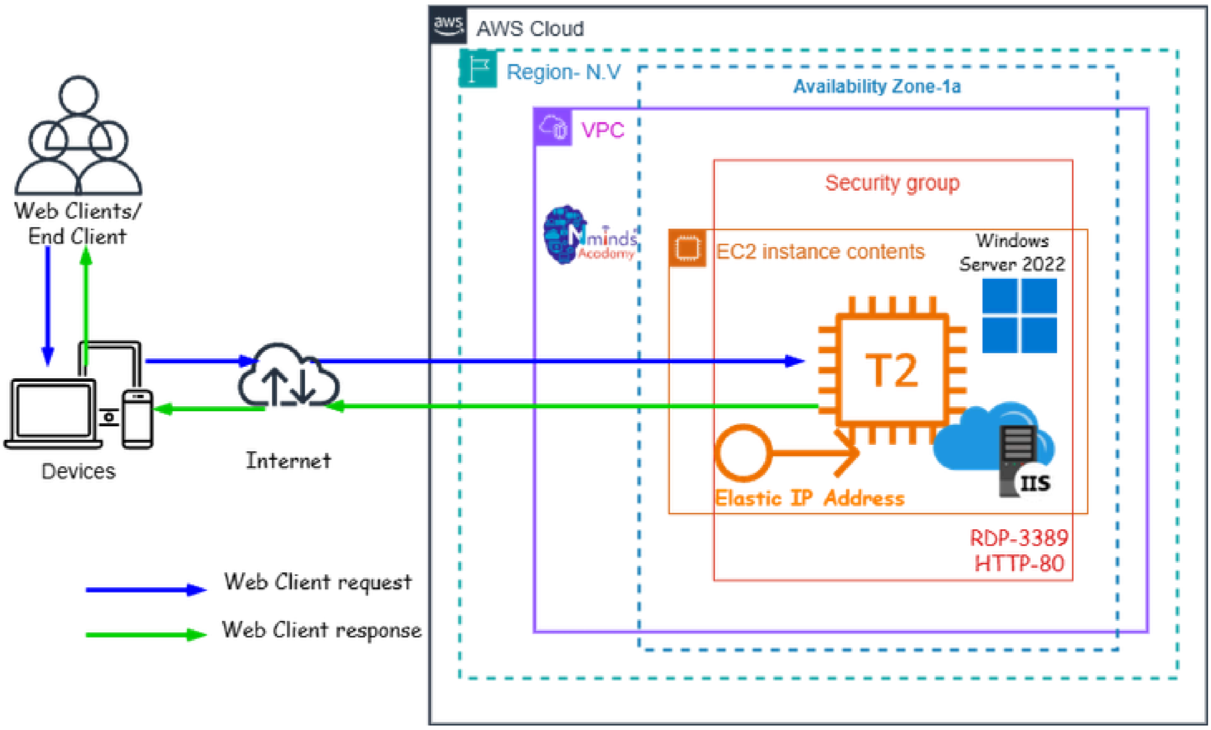
1. Highly Available Applications:
   - o If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.
2. Web Servers:
   - o If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.
3. Disaster Recovery:
   - o Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

Best Practices:

- o Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- o Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- o Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.

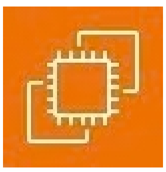
Topology

How to Configure the Elastic IP Address to Windows Web Server with AWS EC2



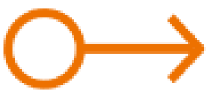
Used Services in AWS

Used Services in Windows Server



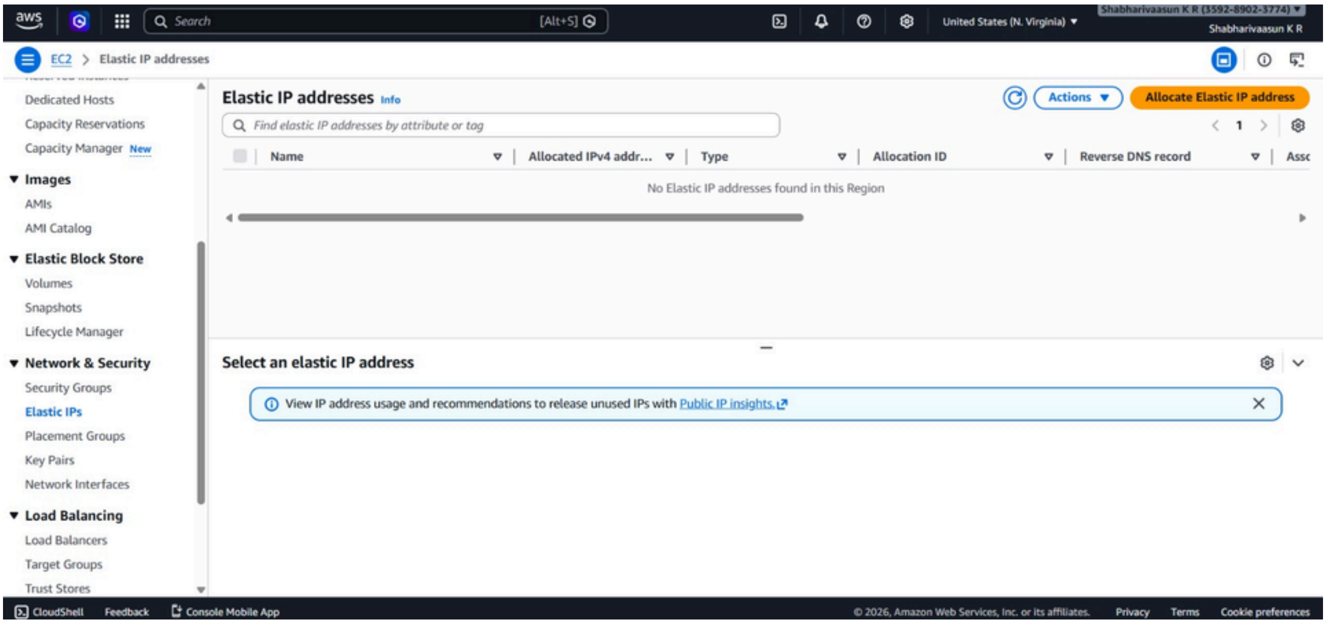
Amazon EC2

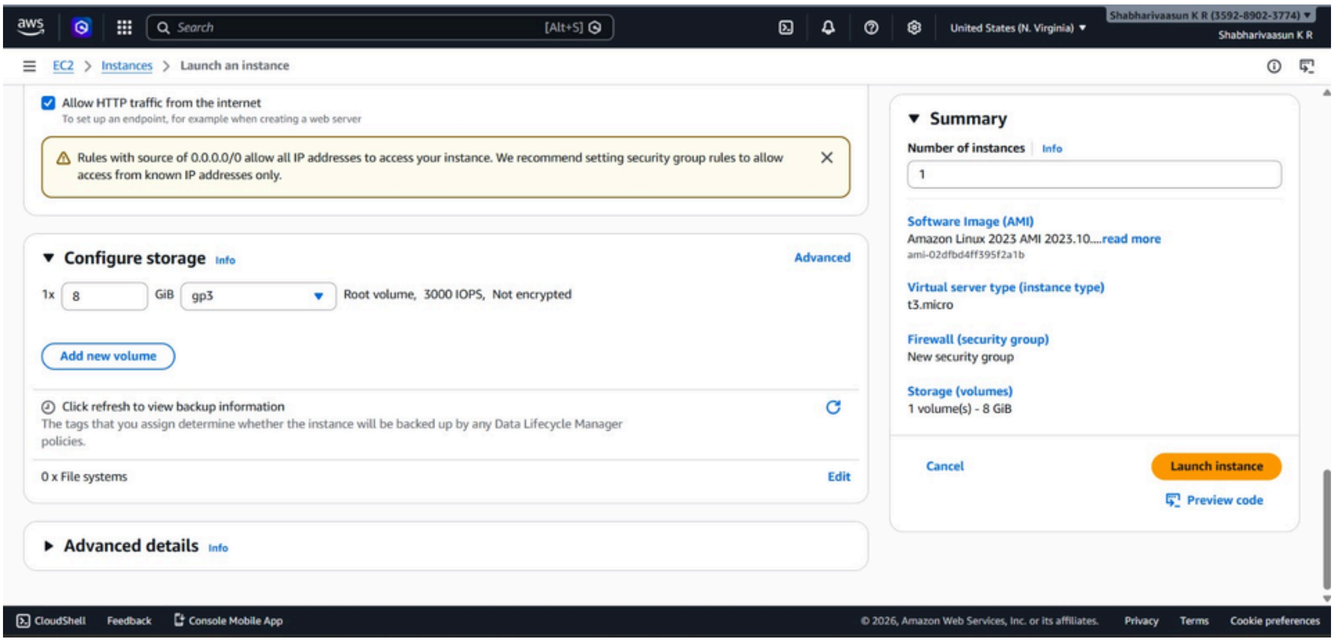
IIS- Internet Information Services



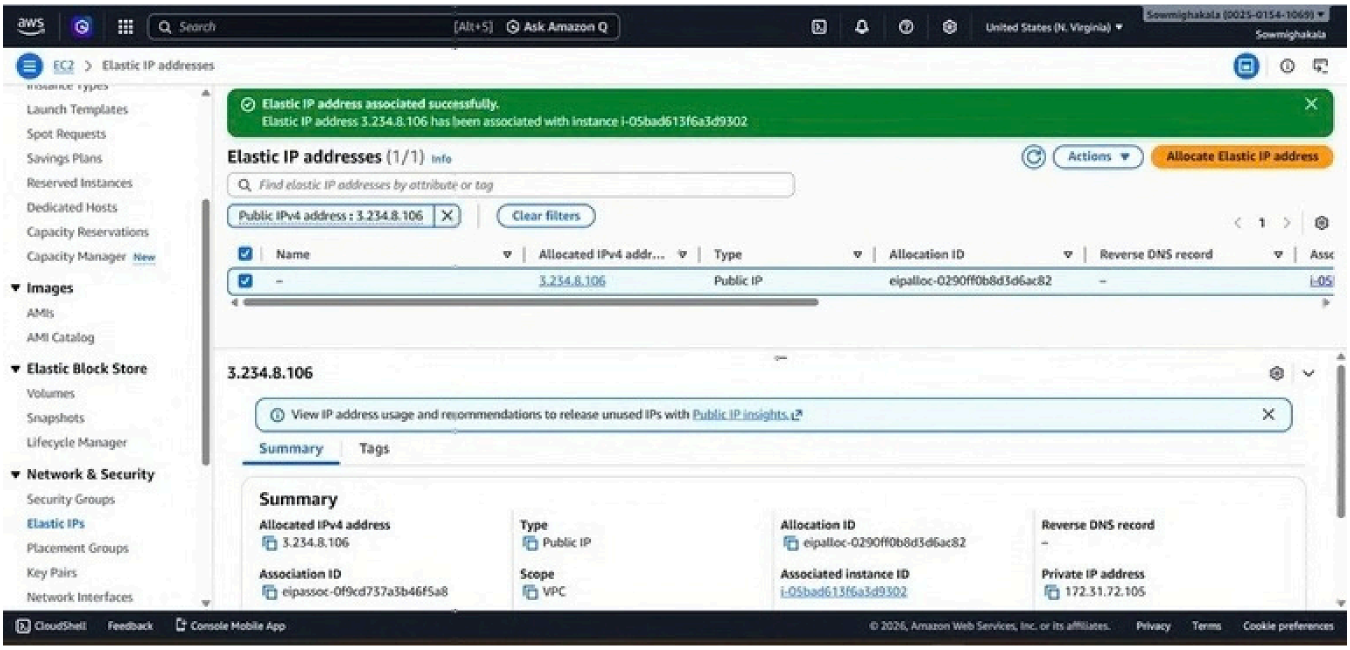
Amazon Elastic IP Address

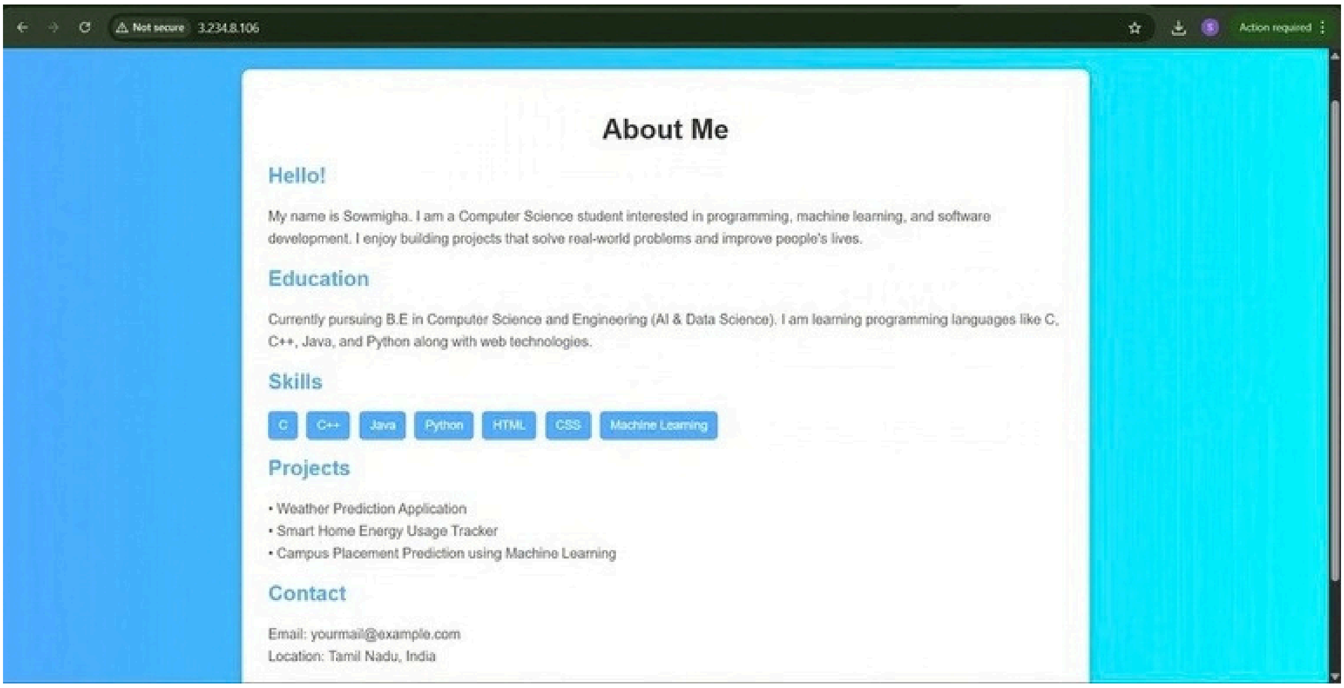
Execution Tasks:





Execution Tasks:





AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

4.Create an EC2 Linux Instance in AWS

SHABHARIVAASUN K R

CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,
Coimbatore.

Objective

EC2 (Elastic Compute Cloud) is a core serviceprovided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache,FileServer etc.) in the cloud without having to manage physical hardware.

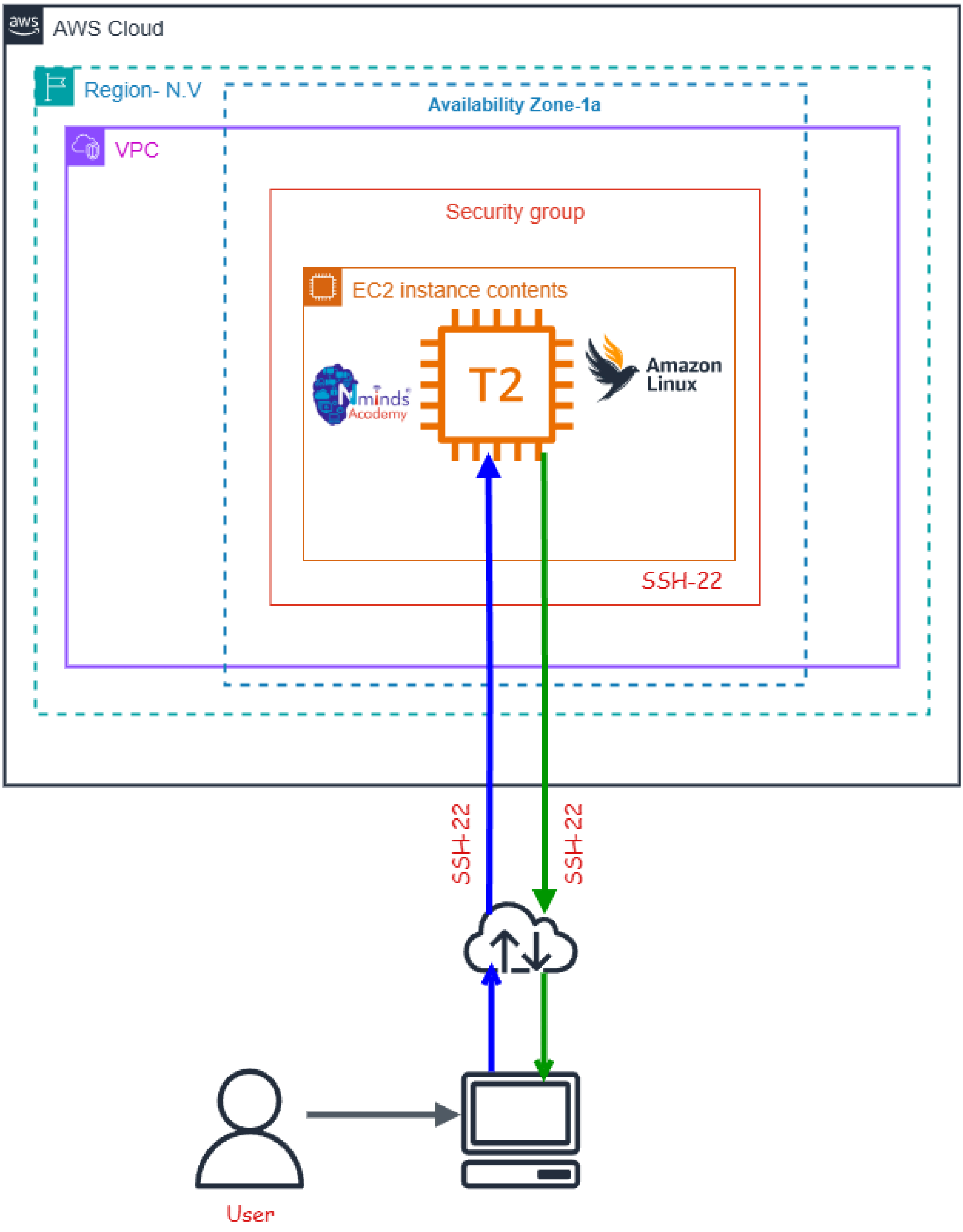
Use Cases for EC2 Linux Instances:

- Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

Used Services in AWS

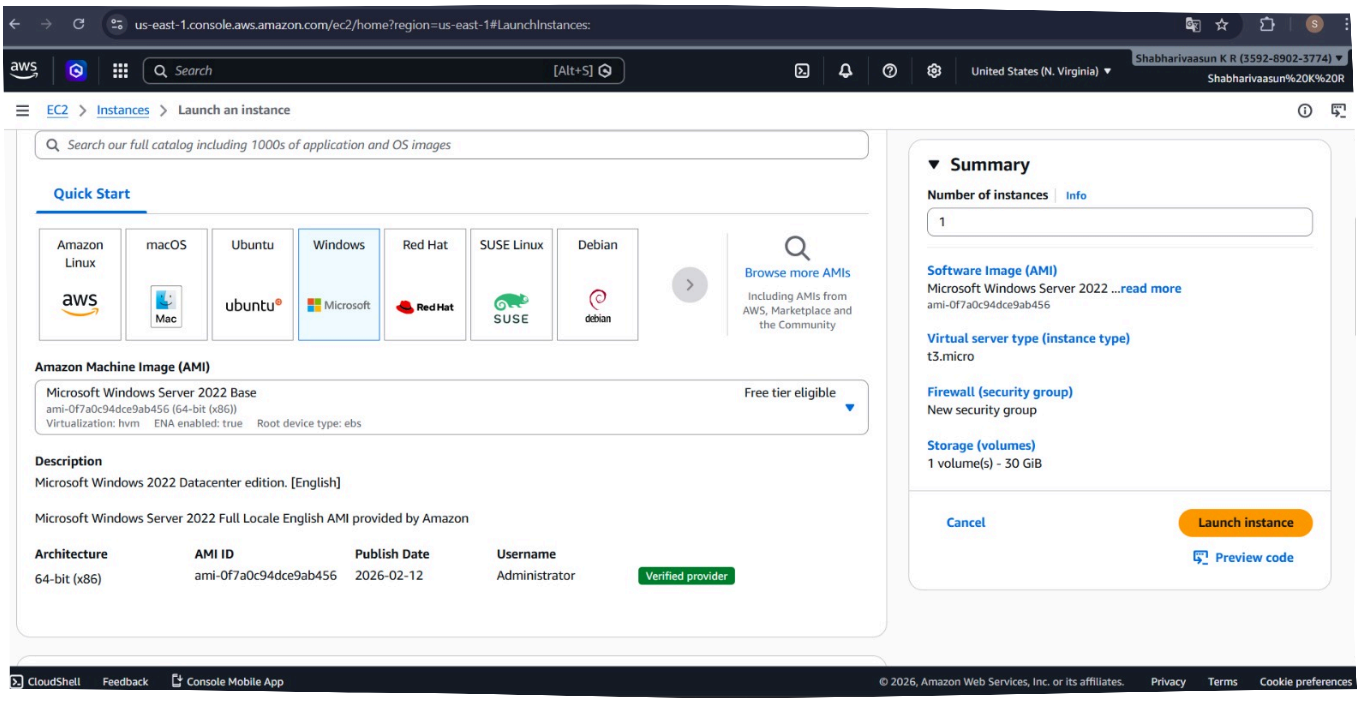
Amazon EC2

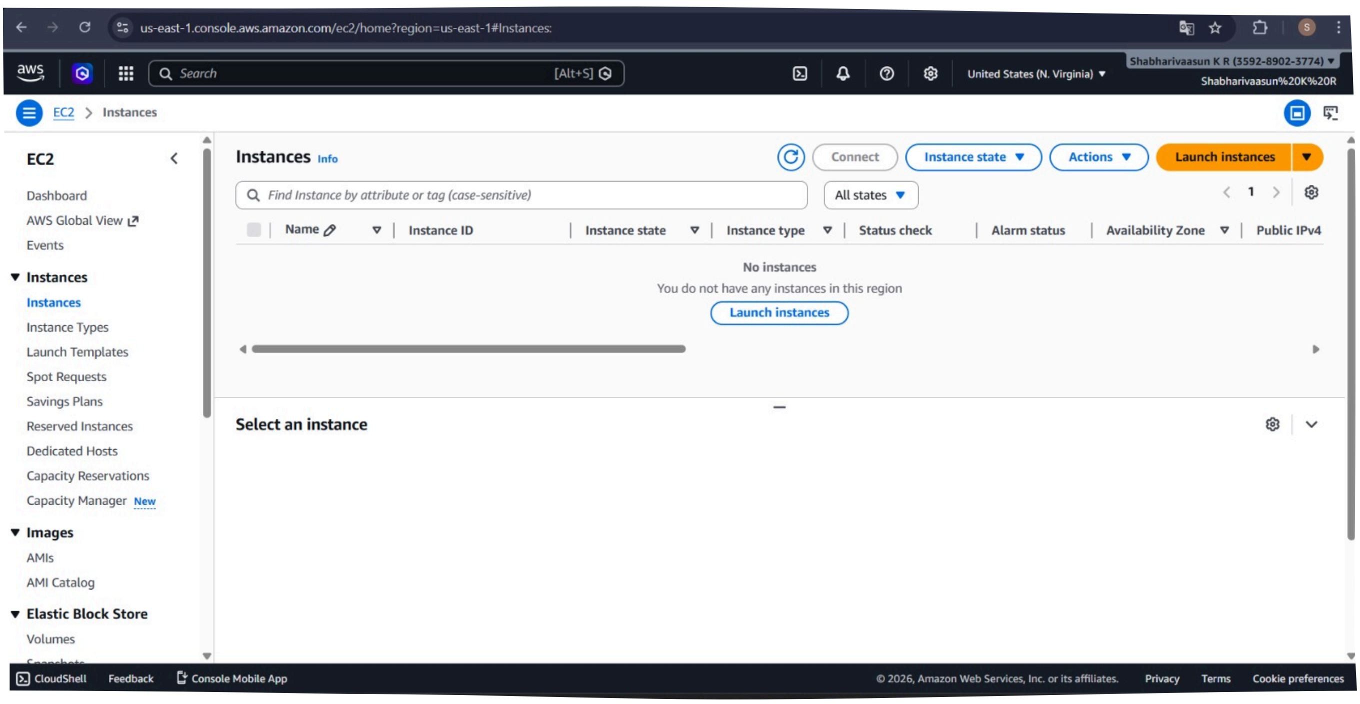
Topology



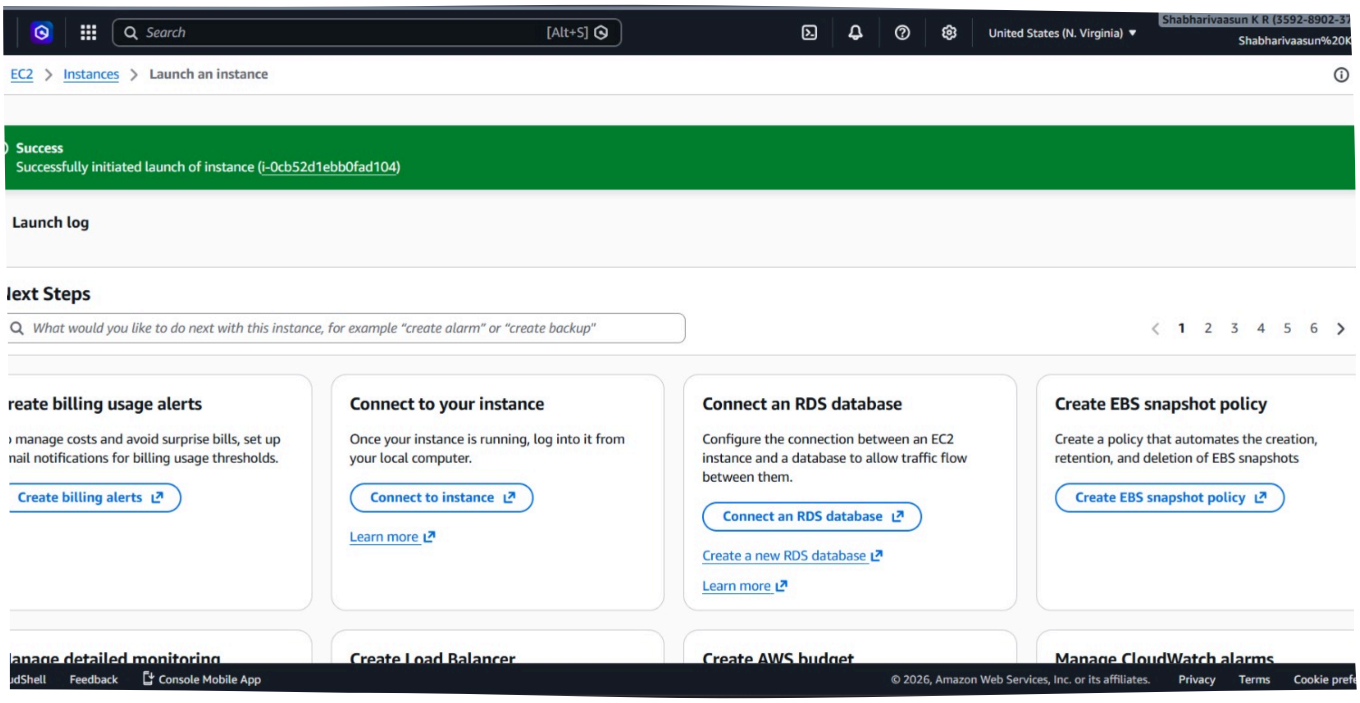
AWS Cloud
Region- N.V
Availability Zone-1a
VPC
Security group
EC2 instance contents
T2
Amazon Linux
Nminds Academy
SSH-22
SSH-22
SSH-22
User

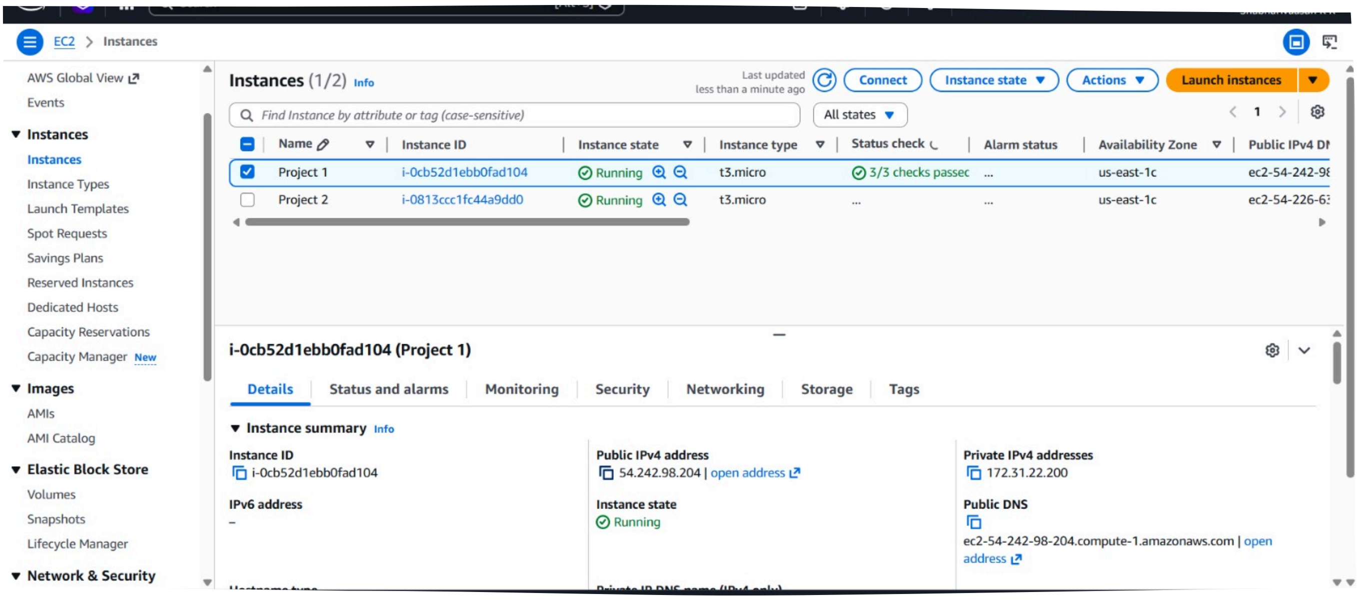
Execution Tasks:





Execution Tasks:





AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

5.Configure Apache Web Server on Linu Instance in x AWS

SHABHARIVAASUN K R

CSE-Artificial Intelligence & Machine Learning,
Sri Eshwar College of Engineering,
Coimbatore.

Objective

EC2 (Elastic Compute Cloud) is a core serviceprovided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache,FileServer etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Linux Instances:

- Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

Used Services in AWS

Amazon EC2

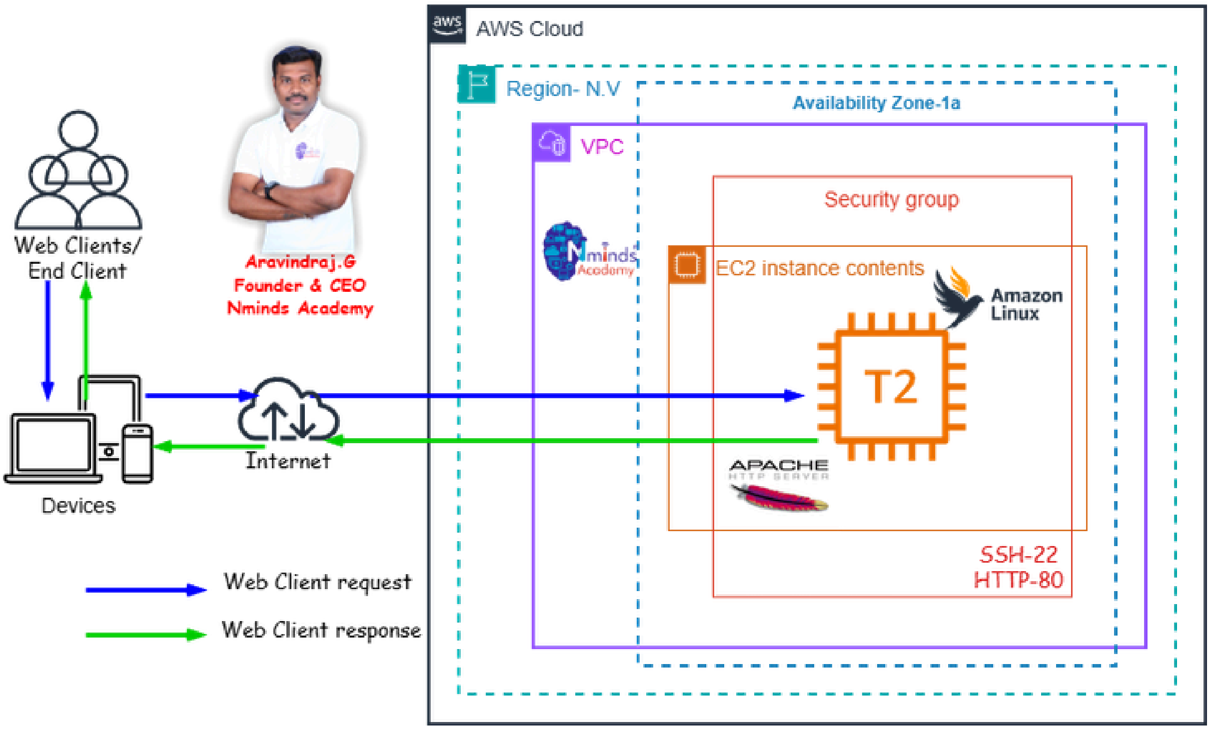
Used Services in Linux Server

APACHE HTTP SERVER

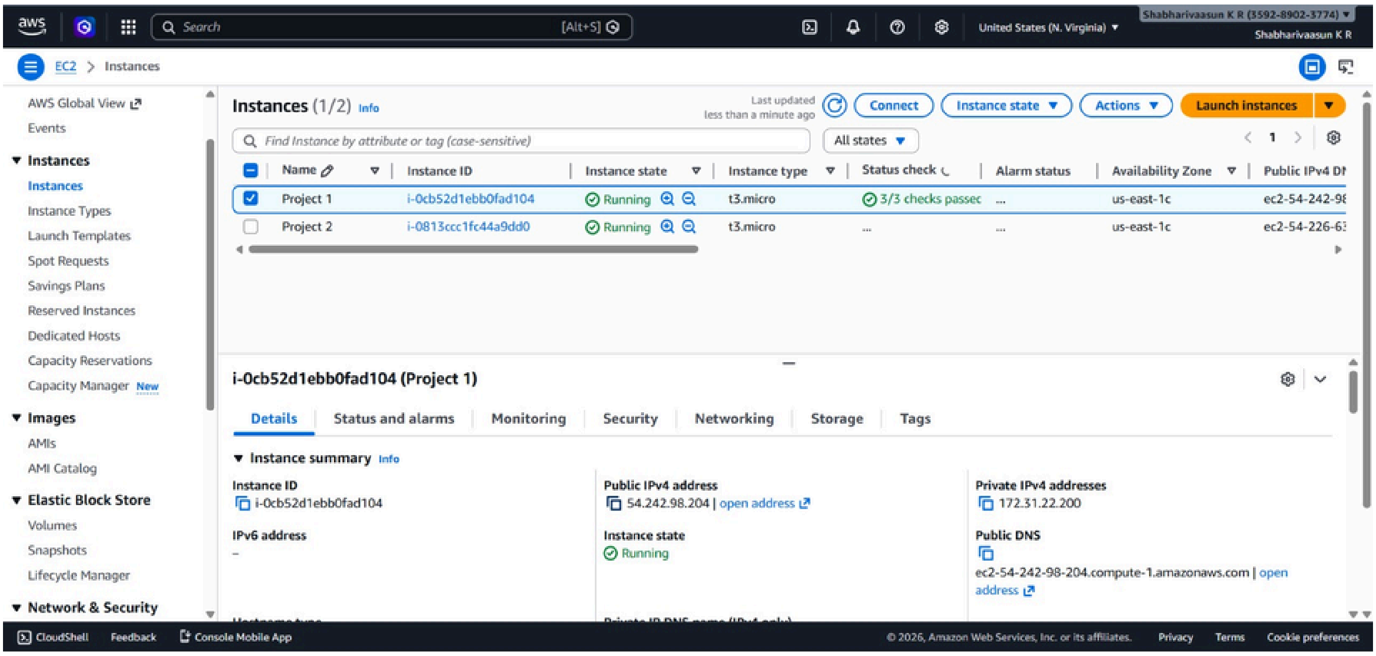
Apache Webserver

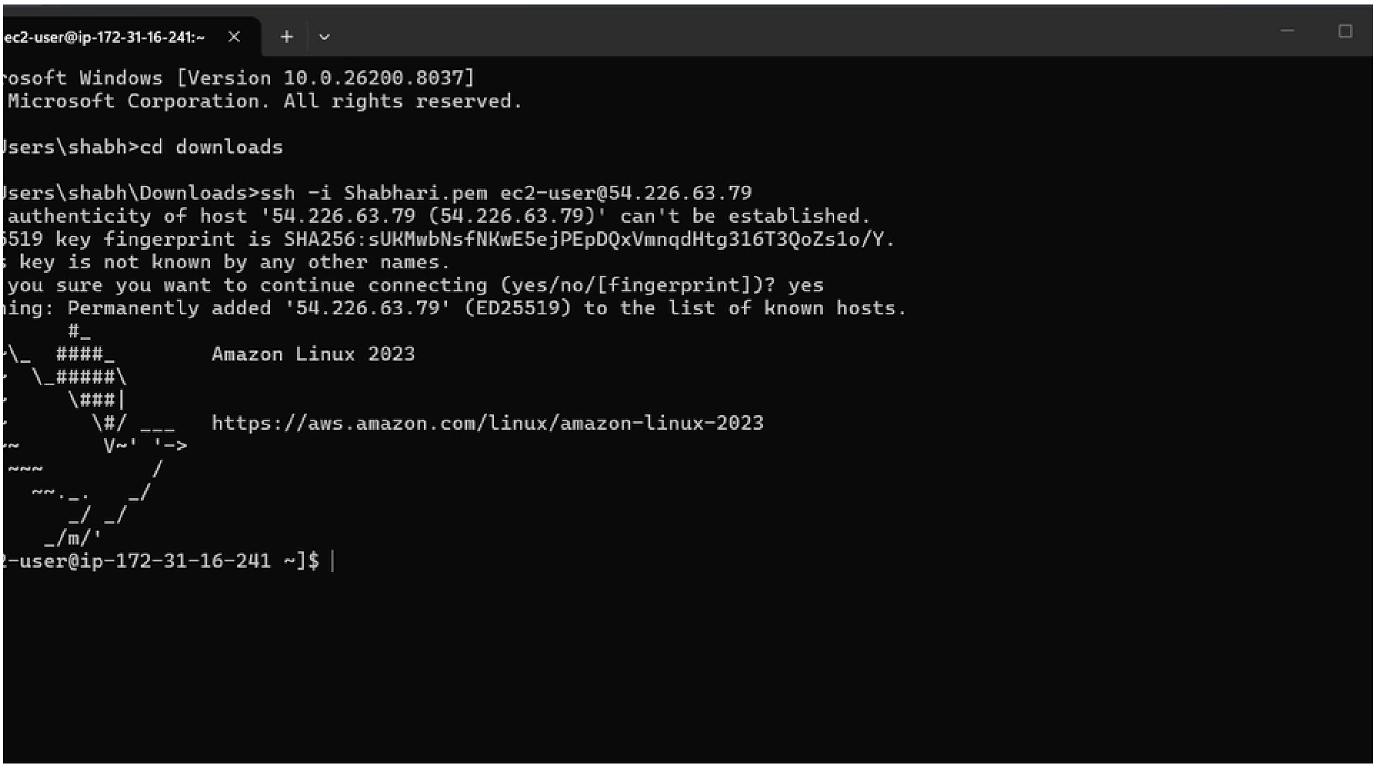
Topology

How to Configure the Apache web server in Linux Server with AWS EC2



Execution Tasks:





Execution Tasks:

```
root@ip-172-31-16-241:~
  Installing       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                 10/13
  Installing       : mod_lua-2.4.66-1.amzn2023.0.1.x86_64                   11/13
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      12/13
  Installing       : httpd-2.4.66-1.amzn2023.0.1.x86_64                     13/13
  Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64                     13/13
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                         1/13
  Verifying        : apr-util-1.6.3-1.amzn2023.0.2.x86_64                    2/13
  Verifying        : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64               3/13
  Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64            4/13
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch       5/13
  Verifying        : httpd-2.4.66-1.amzn2023.0.1.x86_64                      6/13
  Verifying        : httpd-core-2.4.66-1.amzn2023.0.1.x86_64                 7/13
  Verifying        : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch           8/13
  Verifying        : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                9/13
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                  10/13
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                   11/13
  Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                 12/13
  Verifying        : mod_lua-2.4.66-1.amzn2023.0.1.x86_64                   13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64                     apr-util-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64           apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch   httpd-2.4.66-1.amzn2023.0.1.x86_64
  httpd-core-2.4.66-1.amzn2023.0.1.x86_64             httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
  httpd-tools-2.4.66-1.amzn2023.0.1.x86_64            libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch                mod_http2-2.0.27-1.amzn2023.0.3.x86_64
  mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[root@ip-172-31-16-241 ~]#
```

```
root@ip-172-31-72-105:~
# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication no

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes

# Kerberos options
#KerberosAuthentication no
-- INSERT --                                                    65,27         34%
```

Execution Tasks:

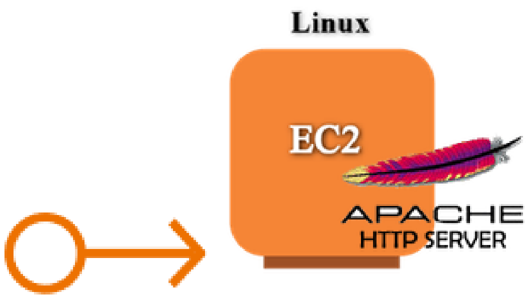
AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

6.Configure Elastic IP Address to Windows Web Server in AWS



SHABHARIVAASUN K R

CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,

Coimbatore.

Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

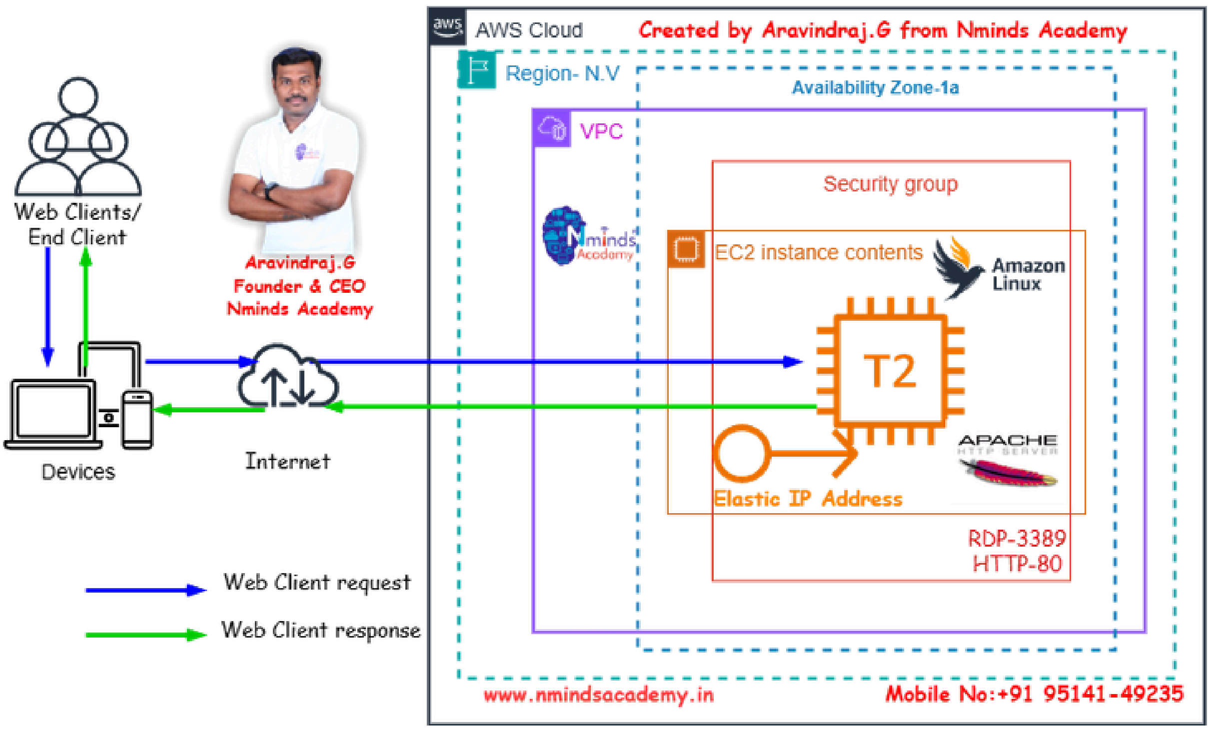
1. Highly Available Applications:
   - o If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.
2. Web Servers:
   - o If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.
3. Disaster Recovery:
   - o Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

Best Practices:

- o Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- o Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- o Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.

Topology

How to Configure the Elastic IP Address to Linux Web Server with AWS EC2



Used Services in AWS
Services in Windows Server

Used

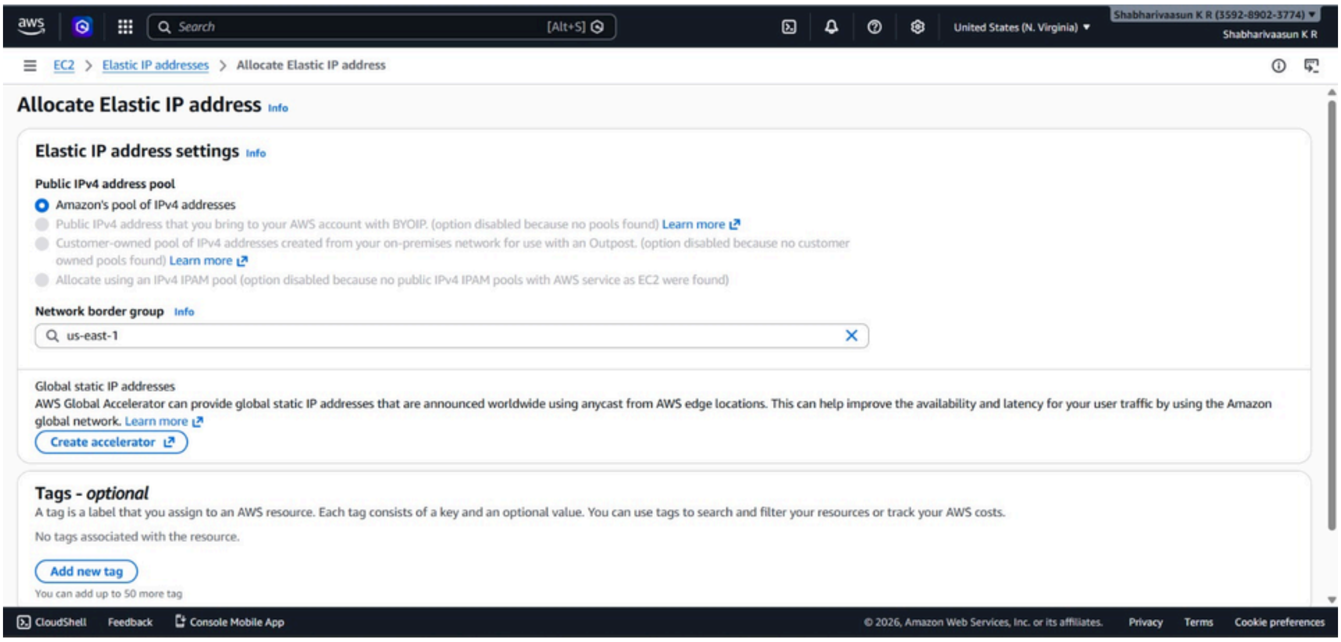
APACHE
HTTP SERVER

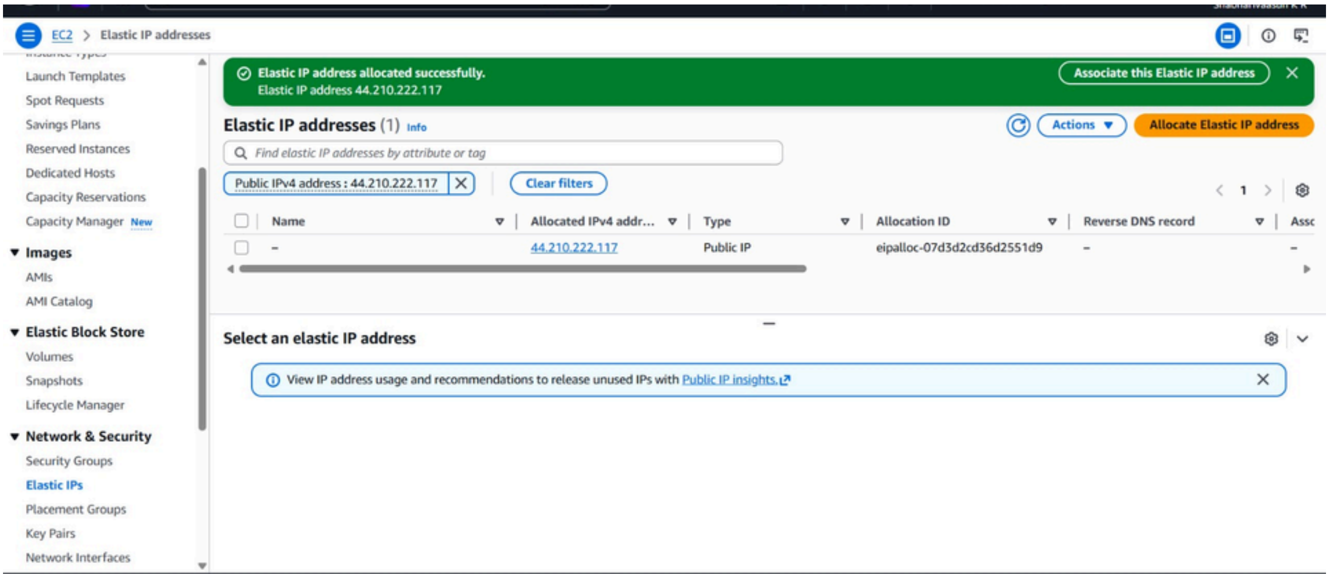
Apache Webserver

Amazon EC2

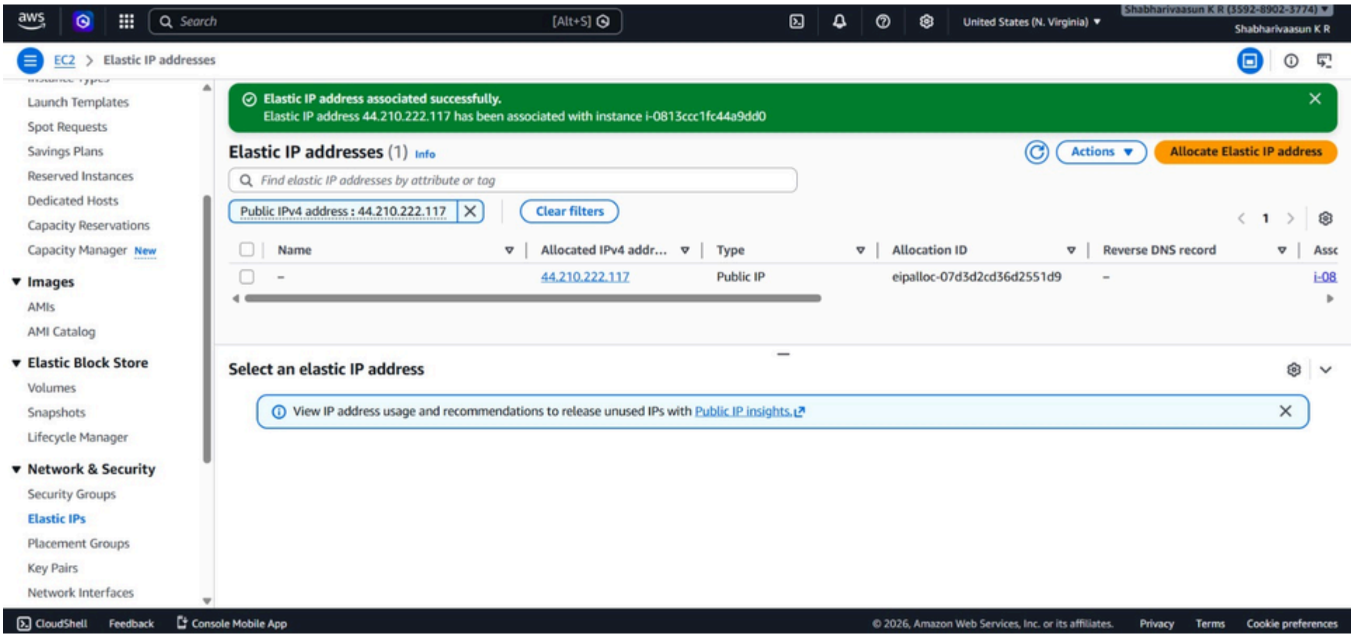
Amazon ElasticIPAddress

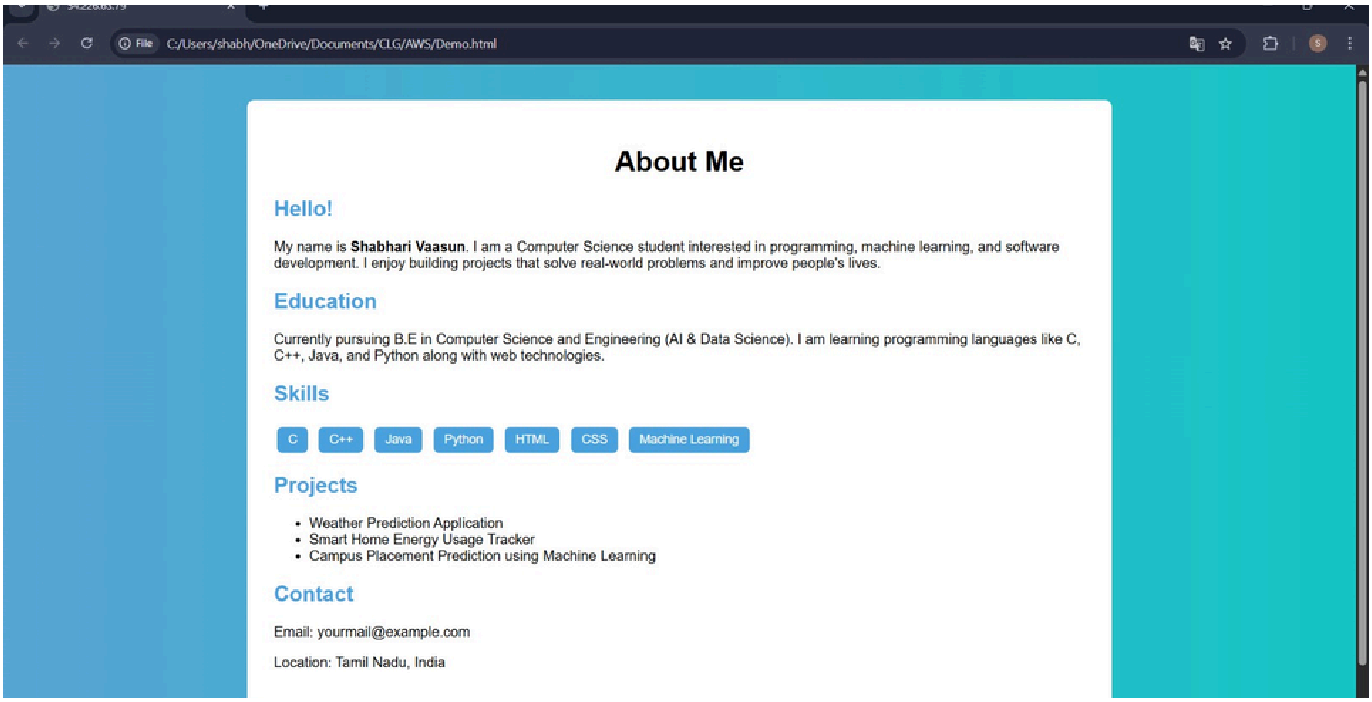
Execution Tasks:





Execution Tasks:





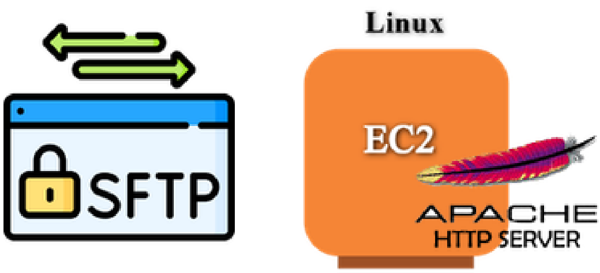
AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

7.Configure SFTP toLinux Web Server in AWS



SHABHARIVAASUN K R

CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,

Coimbatore.

Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

Use Cases for EC2 Linux Instances:

- Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

SFTP- Secure File Transfer Protocol

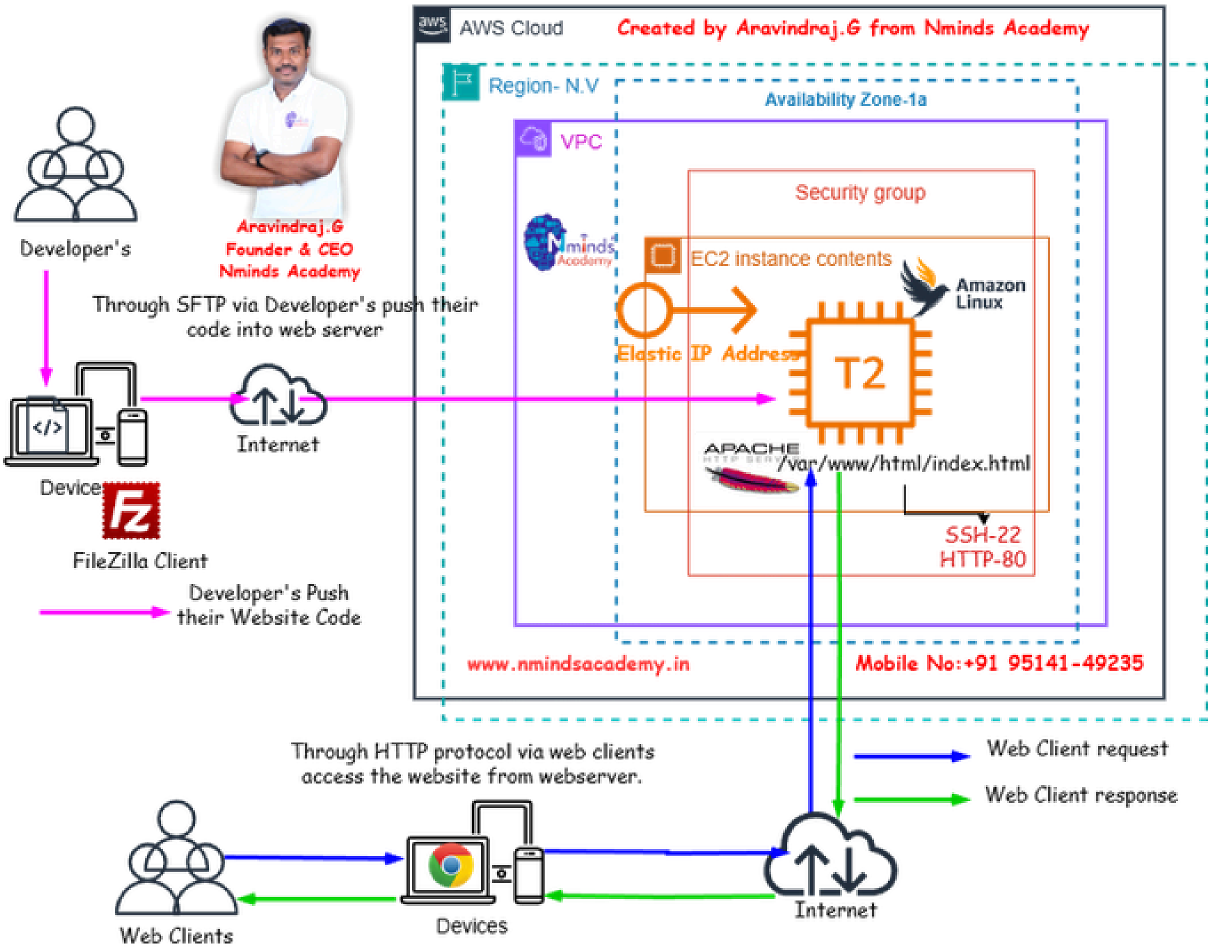
SFTP (Secure File Transfer Protocol) is a secure network protocol that allows users to transfer files over a secure, encrypted connection. It is a more secure alternative to FTP (File Transfer Protocol) and is commonly used to upload, download, and manage files on remote servers.

Here are the key uses and advantages of SFTP:

1. File Transfer: SFTP allows secure transfer of files between a local and a remote computer over a network. It ensures that the data is encrypted during transmission, protecting it from eavesdropping or tampering.
2. File Management: Besides transferring files, SFTP also allows users to perform file management tasks such as renaming, deleting, and moving files on the remote server.
3. Secure Authentication: SFTP uses SSH (Secure Shell) for authentication, which provides a stronger level of security compared to FTP's traditional username/password system. It often supports public key authentication, adding an extra layer of security.
4. Encryption: The main advantage of SFTP over FTP is encryption. All data sent over SFTP is encrypted, ensuring that sensitive information (like passwords and files) is protected during transmission.
5. Firewall-Friendly: SFTP operates over a single port (usually port 22), making it easier to use behind firewalls and NAT (Network Address Translation) devices compared to FTP, which uses multiple ports.

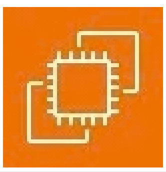
Topology

How to Configure Developer's push their website code by using SFTP Protocol in Linux Server with AWS EC2 with Elastic IP Address



Used Services in AWS

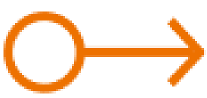
Used Services in Windows Server



Amazon EC2

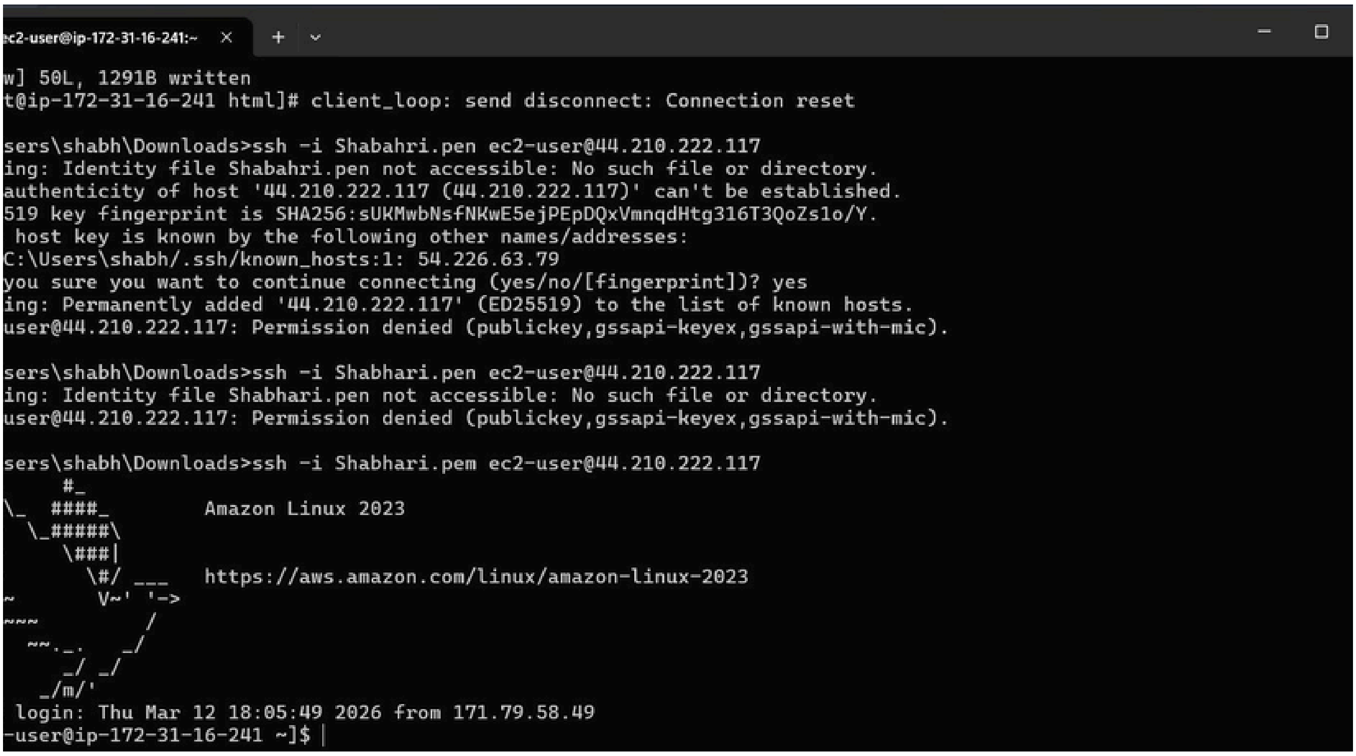
APACHE HTTP SERVER

Apache Webserver

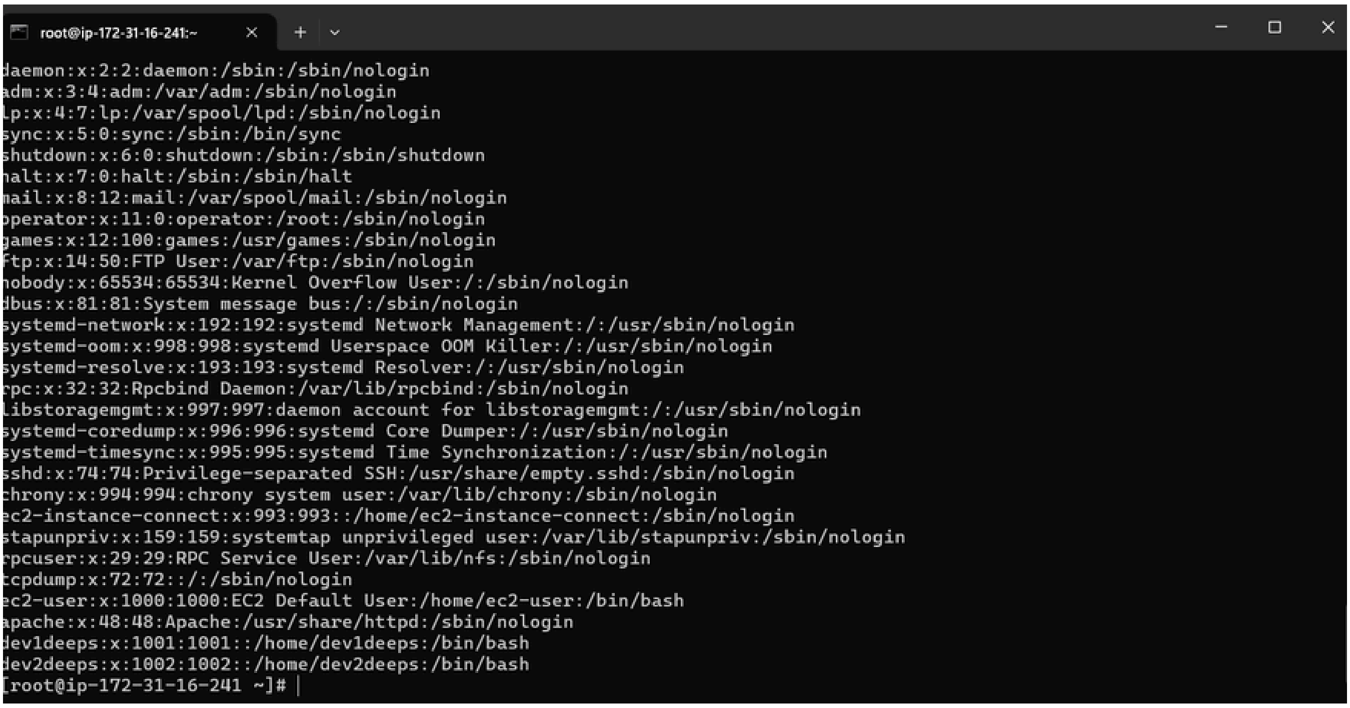


Amazon Elastic IP Address

Execution Tasks:

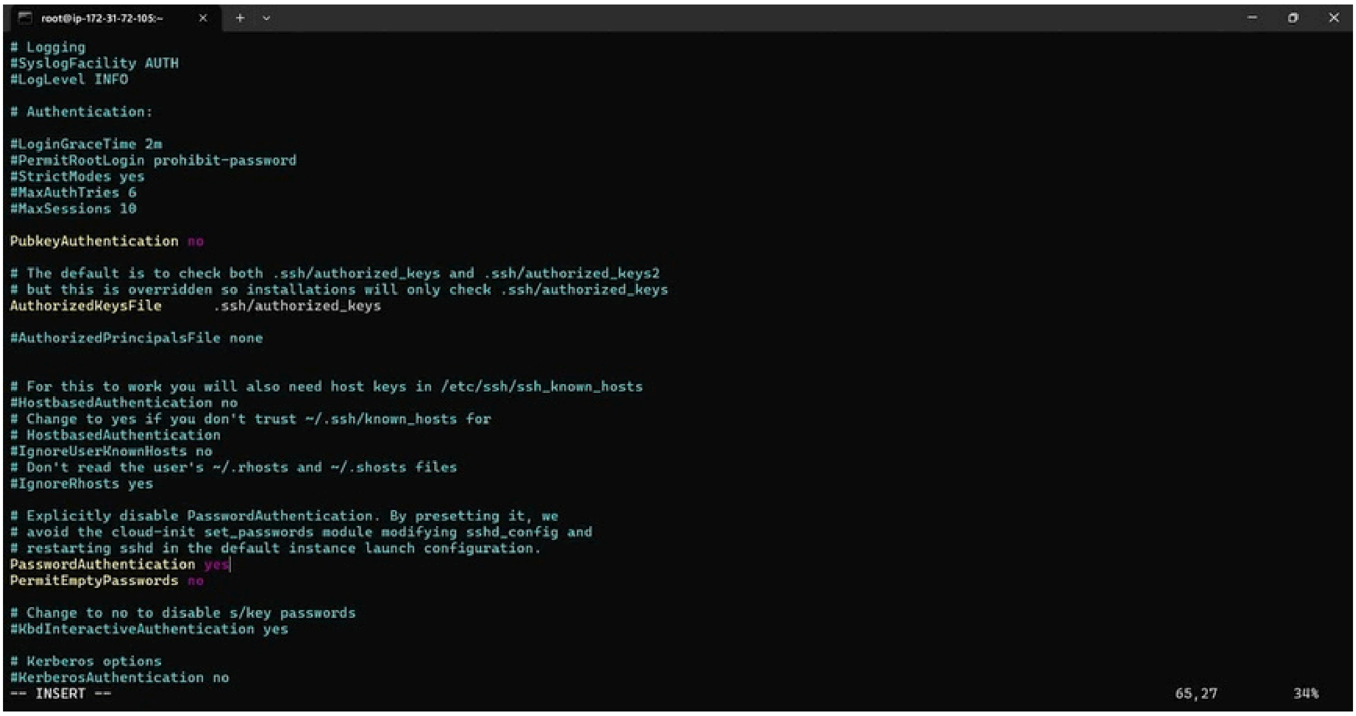


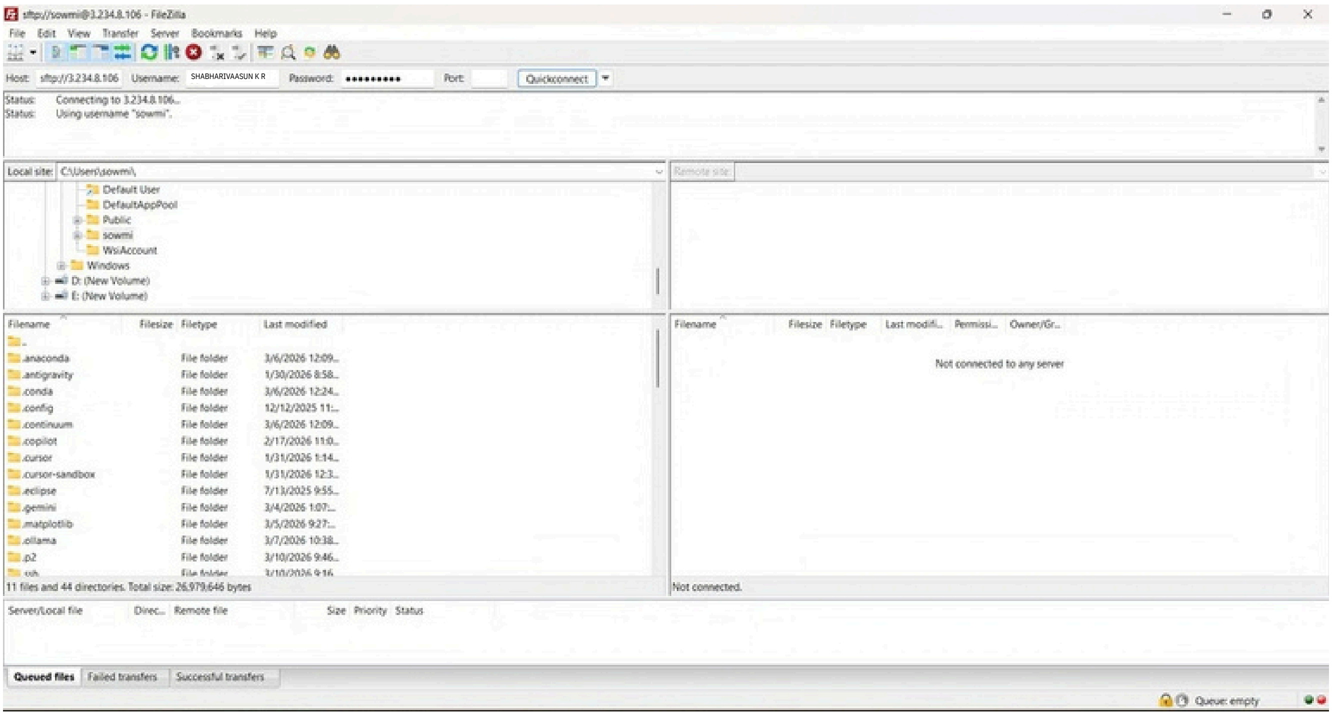
```
ec2-user@ip-172-31-16-241:~
w] 50L, 1291B written
t@ip-172-31-16-241 html]# client_loop: send disconnect: Connection reset

sers\shabh\Downloads>ssh -i Shabahri.pen ec2-user@44.210.222.117
ing: Identity file Shabahri.pen not accessible: No such file or directory.
authenticity of host '44.210.222.117 (44.210.222.117)' can't be established.
519 key fingerprint is SHA256:sUKMwbNsfNKwE5ejPEpDQxVmnqdHtg316T3QoZs1o/Y.
 host key is known by the following other names/addresses:
C:\Users\shabh/.ssh/known_hosts:1: 54.226.63.79
you sure you want to continue connecting (yes/no/[fingerprint])? yes
ing: Permanently added '44.210.222.117' (ED25519) to the list of known hosts.
user@44.210.222.117: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).

sers\shabh\Downloads>ssh -i Shabhari.pen ec2-user@44.210.222.117
ing: Identity file Shabhari.pen not accessible: No such file or directory.
user@44.210.222.117: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).

sers\shabh\Downloads>ssh -i Shabhari.pem ec2-user@44.210.222.117
     #_
\_  ####_        Amazon Linux 2023
  \_#####\
     \###|
       \#/  ___   https://aws.amazon.com/linux/amazon-linux-2023
        V~' '->
~
~~~         /
  ~~._.   _/
     _/ _/
   _/m/'
 login: Thu Mar 12 18:05:49 2026 from 171.79.58.49
-user@ip-172-31-16-241 ~]$ |
```



```
root@ip-172-31-16-241:~
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-oom:x:998:998:systemd Userspace OOM Killer:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/:/usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/:/usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/:/usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin
ec2-instance-connect:x:993:993::/home/ec2-instance-connect:/sbin/nologin
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
dev1deeps:x:1001:1001::/home/dev1deeps:/bin/bash
dev2deeps:x:1002:1002::/home/dev2deeps:/bin/bash
[root@ip-172-31-16-241 ~]# |
```

Execution Tasks:





Execution Tasks: